

Vacation Work Experience at South African Breweries AB InBev

2108498

School of Electrical & Information Engineering, University of the Witwatersrand, Private Bag 3, 2050, Johannesburg, South Africa

Abstract: This report presents an overview of the vacation work completed at South African Breweries (SAB) from December 2024 to February 2025. The objective of the vacation work is to gain practical in industry exposure as an engineer. The work encompassed tasks such as instrumentation, designing and commissioning of electrical infrastructure, completing site investigations into electrical anomalies, assessing site safety, and troubleshooting supervisory control and data acquisition systems (SCADA) systems. This report outlines the technical tasks performed, ethical considerations, time management, and the broader social and economic impact of the work. The experience significantly contributed to my understanding of industrial automation, process optimisation, and system maintenance within a large-scale manufacturing environment.

Supervisor details:

Name:	<u>Tafadzwa Raiva</u>	Accreditation:	<u>ECSA Professional Certificated Engineer,</u>
Position:	<u>Technical Services Manager</u>		<u>MEng. Mechanical Engineering, GCC</u>
Contact:	<u>Tafadzwa.Raiva@za.ab-inbev.com</u>	Signature:	<u></u>

1. INTRODUCTION

The vacation work conducted at South African Breweries (SAB) encompassed various responsibilities where I had been involved in multiple engineering roles and departments. The primary focus was on instrumentation, electrical, and process & control engineering, covering tasks such as instrument verification, electrical design, power usage investigations, and troubleshooting Programmable Logic Controllers (PLC) and SCADA systems. This report details the experience gained and the skills developed during the training period.

2. COMPANY OVERVIEW

SAB is an alcoholic beverage manufacturer who is headquartered in Johannesburg. They are a direct subsidiary of the global brand of Anheuser-Busch (AB InBev). SAB primarily manufactures various brands of beer such as Castle and Black Label as well as other varieties of alcoholic beverages. SAB integrates advanced automation and control systems to facilitate large scale manufacturing in a Fast-Moving Consumer Goods (FMCG) environment. SAB has multiple breweries located throughout Africa.

The vacation work discussed herein was completed at the brewery located in Chamdor where I worked under Technical Services manager Tafadzwa Raiva as an intern. I spent time working in the Utilities and Manufacturing Systems departments. These departments are vital to the functioning of the plant as they service the plants auxiliary functions, supporting the brewing and packaging departments.

3. WORK EXPERIENCE AND RESPONSIBILITIES

3.1 Instrumentation Engineering

The initial phase of the vacation work focused on instrumentation, where the sensors and actuators of the beer filling machines have been verified and maintenance records checked. This aided in learning the manufacturing processes as well as the control systems used. The tasks performed included:

- Calibration of oxygen (O₂) meters to ensure accurate readings for product quality and preservation.
- Inspection and testing of flow sensors and valves to ensure precise filling and minimise inconsistencies.
- Leak detection to identify product losses and maximise yield.

These activities were crucial in maintaining process accuracy and preventing spoilage or contamination in the final product. Working in a production setting highlighted the importance of precise instrumentation to minimise waste and maximise efficiency, reinforcing my technical understanding of industrial measurement systems.

3.2 Electrical Engineering

As part of my electrical engineering responsibilities, I contributed to the design and implementation of an electrical clean room. This involved:

- Managing the construction process by coordinating with contractors and ensuring compliance with required specifications.
- Ensuring environmental control measures such as dust filtration and temperature control were properly implemented.
- Ensuring electrostatic discharge protection for sensitive electronic components.

Additionally, I participated in an investigation into high power consumption on site. This involved data collection, load analysis, and identifying areas where energy efficiency improvements could be implemented. A separate project involved troubleshooting issues related to the Cleaning in Progress (CIP) system. Identifying electrical faults causing manufacturing down time. These experiences provided an opportunity to relate electrical engineering concepts to practical applications with in a FMCG.

3.3 Process and Control Engineering

My exposure to process and control engineering included:

- Gaining an understanding of SAB's SCADA architecture and its role in plant automation.
- Working with Schneider and Krones control systems to analyse and oversee process automation.
- Assisting in troubleshooting PLC, motor driver, and network connection issues to minimise downtime.
- Collaborating with maintenance teams to diagnose and resolve faults in control loops and automation networks.

3.4 Safety Compliance

Beyond technical tasks, I also contributed to safety compliance projects:

- Elevator Lift Safety Assessment: I assisted in evaluating the condition of elevator lifts on site and facilitated the process of registering them with the government, ensuring compliance with legal safety standards.
- Fire Safety Improvements: I worked with fire safety specialists to identify inadequate alarm coverage and improve fire detection systems, improving overall site safety.

These experiences contributed to both regulatory compliance and overall workplace safety at the facility. The diverse nature of my responsibilities enabled me to develop a well-rounded understanding of large-scale industrial engineering, combining both technical problem-solving and project management aspects. The experience of managing contractors, coordinating with safety specialists, and troubleshooting complex systems has furthered my ability to work in a professional engineering environment.

4. TIME MANAGEMENT

The workday at SAB began at 8h00 with a safety meeting in the morning and ends at 16h30. A lunch break is taken between 12h00 and 13h00. Detailed below is a week-by-week breakdown of the work completed. A breakdown of the daily tasks is detailed in the Vacation Work diary attached in Appendix B at the end of this document. Moreover, an assessment certificate of the vacation work completed is also attached in Appendix B.

Week 1-2: Instrumentation and Initial Assessments

During the first two weeks, I focused on instrument verification. This included calibrating oxygen meters, inspecting flow sensors, and performing leak detection tests on beer filling machines and piping. I also familiarised myself

with the plant layout, safety protocols, and maintenance schedules. Moreover, I performed my safety inductions required to be on site. Figures of the aforementioned sensors and filling machines are attached below in Appendix A.

Week 3-4: Electrical Engineering and Clean Room Design

I worked on designing and implementing an electrical clean room, managing the construction process by coordinating with contractors and ensuring compliance with environmental control measures. Additionally, I began an investigation into the plant's high-power usage, analysing data logs and identifying potential inefficiencies in energy consumption. Figures displaying backup power generators are attached in Appendix A

Week 5-6: CIP System Investigation and Power Consumption Analysis

Midway through the program, I conducted an in-depth study on the Cleaning in Progress (CIP) system, focusing on electrical faults causing inefficiencies. I also continued my work on the plant's power consumption analysis, recommending energy-saving solutions based on my findings.

Week 7-8: SCADA and Process Control Troubleshooting

During this period, I gained hands-on experience with SAB's SCADA architecture, working with Schneider and Krones systems. I assisted in troubleshooting PLC and network-related issues, collaborating with maintenance teams to minimise downtime and ensure production continuity. Figures of the aforementioned PLC infrastructure are attached below in Appendix A

Week 9-10: Safety and Compliance Projects

I contributed to site-wide safety improvements, assisting with the assessment and government registration of elevator lifts. Additionally, I worked with fire safety specialists to improve the plant's alarm systems by identifying and addressing areas with inadequate coverage.

Week 11-12: Final Assessments and Project Wrap-Up

In the final two weeks, I worked on a part-time basis and provided general support across various departments. My tasks included minor troubleshooting, assisting with documentation, and observing plant operations to gain a broader understanding of industrial processes. This period allowed me to reflect on my experiences and consolidate my knowledge in preparation for future engineering roles.

5. PROFESSIONAL ETHICS

Engineering ethics formed an important aspect in my work at SAB, influencing decisions related to safety, accuracy, teamwork, and sustainability. Adhering to strict workplace safety regulations was essential, particularly in high-risk environments involving electrical systems and automated machinery. Lockout/tagout (LOTO) procedures and adherence to industry standards ensured that maintenance and troubleshooting tasks were conducted safely.

Accuracy and integrity in instrumentation calibration were vital in maintaining product quality and production efficiency. Any errors could lead to significant financial and operational setbacks. I also worked closely with engineers, technicians, and contractors, emphasising transparent communication and collaboration to successfully execute projects such as the electrical clean room and fire safety improvements.

Environmental responsibility was another key aspect of my work. The power consumption analysis and process control improvements aimed at reducing energy waste and promoting sustainability, aligning with SAB's commitment to resource efficiency and cost reduction.

6. SOCIAL IMPACT

6.1 Benefits to Society

The work I performed had direct benefits for both SAB and the broader community. Ensuring accurate instrumentation calibration contributed to consistent product quality and minimised waste, which benefits consumers and reduces production inefficiencies. My involvement in power consumption analysis helped identify ways to lower energy usage, contributing to environmental sustainability. Additionally, safety initiatives such as elevator lift assessments and fire safety improvements enhanced workplace conditions, protecting employees and ensuring compliance with national safety standards.

6.2 Economic Considerations

The work I performed had direct financial implications, particularly in reducing operational costs. Identifying energy efficiency improvements helped lower power expenses, supporting SAB's long-term sustainability. The electrical clean room project reduced risks related to equipment failures, minimising replacement costs. Fire safety enhancements also demonstrated the value of preventative investment, mitigating potential financial losses due to workplace hazards.

7. CONCLUSIONS

The vacation work experience at SAB provided invaluable exposure to industrial engineering applications, allowing me to develop technical proficiency in instrumentation, electrical systems, and process control. Managing the construction of an electrical clean room, investigating power consumption inefficiencies, and contributing to fire safety improvements reinforced my ability to apply engineering principles in a real-world setting. Collaborating with engineers, technicians, and contractors enhanced my teamwork and problem-solving skills, while adherence to professional ethics ensured safe and responsible engineering practices. The experience deepened my understanding of industrial automation and process efficiency, preparing me for future engineering roles and contributing to operational improvements at SAB in energy conservation, safety, and quality assurance.

Appendix A:



Figure 1a) - Figure displaying filling machine O₂ sensor



Figure 1b) - Figure displaying CO₂ actuator for filling machine



Figure 1c) - Figure displaying filling machine in operation



Figure 2) - Figure displaying backup generators



Figure 3) - Figure displaying air compressors



Figure 4) – Figure displaying filler machine PLC infrastructure



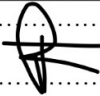
Figure 5) - Figure displaying brewing PLC infrastructure

Appendix B:

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Faculty of Engineering and the Built Environment

Vacation Work Assessment Certificate

This covering sheet must be attached to your detailed Vacation Work Report as the first page of the report. Both the covering sheet and the last page of the Vacation Work Report must be signed by your supervisor/ employer and handed to the School Administrative Officer in your branch **BY THE SUBMISSION DATE ANNOUNCED BY THE SCHOOL.**

STUDENT NAME: Yasteel Ramphal
STUDENT NUMBER: 2108498
EMAIL ADDRESS: yasteel001@gmail.com TELEPHONE NO: 0838746373
YEAR OF STUDY: 4 SCHOOL (and option if applicable) EIE
EMPLOYER'S NAME: South African Breweries AB InBev
EMPLOYER'S ADDRESS: Chamdor Brewery, 3 Fransen Street, Krugersdorp, Gauteng,
South Africa
EMPLOYER'S TELEPHONE NO: 011 951 2600
DATE ENTERED SERVICE: 1 December 2024 DATE LEFT SERVICE: 28 February 2025
OUTLINE OF WORK DONE: Instrumentation, electrical engineering design and investigation
process & control engineering and site safety investigation
HAS THE STUDENT PERFORMED SATISFACTORILY? Yes he performed well during his period
GENERAL CONDUCT: The student did well to identify his learning needs while on site and
work with relevant departments
SUPERVISOR'S/EMPLOYER'S SIGNATURE: 
DATE: 28/02/2025 SUPERVISOR'S/EMPLOYER'S NAME: Tafadzwa Raiva
SUPERVISOR'S/EMPLOYER'S CONTACT NO: 072 234 5612

COMPANY STAMP:

UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG
FACULTY OF ENGINEERING AND THE BUILT ENVIRONMENT

VACATION WORK DIARY FOR ELEC & INFOR ENGINEERING STUDENTS

To be handed in to your Departmental Administrative Officer by **date assigned by the School**.
 (A separate form is required for each organization at which the student works.)

STUDENT NO: 2108498 YEAR OF STUDY: 4

NAME: Yasteel Ramphal

NAME OF EMPLOYER: South African Breweries AB InBev

ADDRESS OF EMPLOYER: Chamdor Brewery, 3 Fransen Street, Krugersdorp, Gauteng, South Africa

DATE ENTERED SERVICE: 1 December 2024 DATE LEFT SERVICE: 28 February 2025

USE ONE LINE FOR EACH SHIFT WORKED (ADD LINES/PAGES AS NEEDED)

	Date	Department	Nature of Work Undertaken
1	02-Dec-2024	Admin	Allocation to Brewery
2	03-Dec-2024	Admin	
3	04-Dec-2024	Admin	
4	05-Dec-2024	Admin	
5	06-Dec-2024	Admin	
6	09-Dec-2024	Utilities	Safety Inductions for site
7	10-Dec-2024	Utilities	Site tour and team introductions
8	11-Dec-2024	Utilities	Assisted and followed machine operators
9	12-Dec-2024	Utilities	Assisted and followed machine operators
10	13-Dec-2024	Utilities	Inspection and verification of flow sensors.
11	16-Dec-2024	Utilities	Data logging and instrument validation of filler 4
12	17-Dec-2024	Utilities	Data logging and instrument validation of filler 4
13	18-Dec-2024	Utilities	Data logging and instrument validation of filler 3
14	19-Dec-2024	Utilities	Data logging and instrument validation of filler 3
15	20-Dec-2024	Utilities	Electrical clean room research and design
16	23-Dec-2024	Utilities	Electrical clean room research and design
17	24-Dec-2024	Utilities	Data logging and instrument validation of filler 7
18	02-Jan-2025	Utilities	Data logging and instrument validation of filler 7
19	03-Jan-2025	Utilities	Documentation of instrumentation
20	06-Jan-2025	Utilities	Electrical clean room research and design
21	07-Jan-2025	Utilities	Documentation of instrumentation
22	08-Jan-2025	Utilities	Electrical clean room contracting
23	09-Jan-2025	Utilities	Electrical lifts inspection
24	10-Jan-2025	Utilities	Electrical lifts inspection
25	13-Jan-2025	Manufacturing Systems	Identified fire safety alarm coverage gaps
26	14-Jan-2025	Manufacturing Systems	Coordinated with fire safety specialists for solutions
27	15-Jan-2025	Manufacturing Systems	Reviewed site-wide energy usage
27	16-Jan-2025	Manufacturing Systems	Electrical clean room contracting
29	17-Jan-2025	Manufacturing Systems	Investigated SCADA systems

30	20-Jan-2025	Manufacturing Systems	Electrical clean room contracting and construction planning
31	21-Jan-2025	Manufacturing Systems	Learing SCADA systems architecture
32	22-Jan-2025	Manufacturing Systems	Site electrical system investigation
33	23-Jan-2025	Manufacturing Systems	Site electrical system investigation
34	27-Jan-2025	Manufacturing Systems	Cleaning In Progress Investigation
36	28-Jan-2025	Manufacturing Systems	Updating electrical safety posters
37	29-Jan-2025	Manufacturing Systems	Troubleshooting PLC errors
38	30-Jan-2025	Manufacturing Systems	Consulting with fire safety specialists
39	31-Jan-2025	Manufacturing Systems	Updating electrical safety posters
40	03-Feb-2025	Manufacturing Systems	Cleaning In Progress Investigation
41	04-Feb-2025	Manufacturing Systems	Shadowed control engineers
42	05-Feb-2025	Manufacturing Systems	Assisted in troubleshooting PLC errors
43	06-Feb-2025	Manufacturing Systems	Assisted in troubleshooting network faults.
44	07-Feb-2025	Manufacturing Systems	Investigation into backup power generation
45	10-Feb-2025	Manufacturing Systems	Creating PLC system logic
46	11-Feb-2025	Manufacturing Systems	Creating PLC system logic
47	12-Feb-2025	Manufacturing Systems	Creating PLC system logic
48	13-Feb-2025	Manufacturing Systems	Assisted SCADA specialists with setting up virtual clients
49	14-Feb-2025	Manufacturing Systems	Followed brand change procedure
50	17-Feb-2025	Manufacturing Systems	Completed minor troubleshooting task
51	21-Feb-2025	Manufacturing Systems	Completed minor troubleshooting tasks
52	24-Feb-2025	Manufacturing Systems	Completed minor troubleshooting tasks
53	28-Feb-2025	Manufacturing Systems	Final day, closing procedure

2024 VACWORK DIARY

Date: 28/02/2025

Signature:  .